

PRACTICAL COURSE WORKBOOK

Health Informatics & Digital Health

Work with digital-health information responsibly

LEVEL	Beginner
GUIDED TIME	5 hours across 12 lessons
PROJECT	a safe, fictional health-data workflow and review
SKILLS	Health IT, EHR, FHIR, HL7, Telemedicine, Critical evaluation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use Health IT appropriately in a realistic, bounded task.
- Use EHR appropriately in a realistic, bounded task.
- Use FHIR appropriately in a realistic, bounded task.
- Use HL7 appropriately in a realistic, bounded task.

WHO THIS IS FOR

Learners exploring health technology without replacing clinical expertise.

BEFORE YOU BEGIN

- Basic data literacy

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

Health Informatics & Digital Health is a structured, practical course covering Health IT, EHR, FHIR, HL7, Telemedicine. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Health IT: Health context	1. Care pathways with Health IT 2. Health data with EHR 3. Stakeholders with FHIR
02 EHR: Digital systems	1. Standards with HL7 2. Interoperability with Telemedicine 3. Workflows with Health IT
03 FHIR: Quality and safety	1. Validation with EHR 2. Privacy with FHIR 3. Human oversight with HL7
04 HL7: Applied evaluation	1. Implementation with Telemedicine 2. Monitoring with Health IT 3. Equity with EHR

HOW TO USE EACH LESSON PAGE

- 1 Read the objective and identify the decision it supports.
- 2 Attempt the retrieval prompt before checking the online explanation.
- 3 Reproduce the worked example with the stated input.
- 4 Test one normal case and one boundary or failure case.
- 5 Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

Health IT: Health context

Apply Health IT through care pathways, health data, stakeholders.

LESSON 01 / 18 MIN

Care pathways with Health IT

Learning objectives

- Explain care pathways in the context of Health Informatics & Digital Health.
- Apply Health IT to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

care pathways, Health IT, healthcare

Retrieval prompt

In Health Informatics & Digital Health, which evidence best supports a care pathways result produced with Health IT?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 02 / 22 MIN

Health data with EHR

Learning objectives

- Explain health data in the context of Health Informatics & Digital Health.
- Apply EHR to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

health data, EHR, healthcare

Retrieval prompt

In Health Informatics & Digital Health, which evidence best supports a health data result produced with EHR?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 03 / 26 MIN

Stakeholders with FHIR

Learning objectives

- Explain stakeholders in the context of Health Informatics & Digital Health.
- Apply FHIR to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

stakeholders, FHIR, healthcare

Retrieval prompt

In Health Informatics & Digital Health, which evidence best supports a stakeholders result produced with FHIR?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

EHR: Digital systems

Apply EHR through standards, interoperability, workflows.

LESSON 04 / 30 MIN

Standards with HL7

Learning objectives

- Explain standards in the context of Health Informatics & Digital Health.
- Apply HL7 to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

standards, HL7, healthcare

Retrieval prompt

In Health Informatics & Digital Health, which evidence best supports a standards result produced with HL7?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 05 / 18 MIN

Interoperability with Telemedicine

Learning objectives

- Explain interoperability in the context of Health Informatics & Digital Health.
- Apply Telemedicine to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

interoperability, Telemedicine, healthcare

Retrieval prompt

In Health Informatics & Digital Health, which evidence best supports a interoperability result produced with Telemedicine?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 06 / 22 MIN

Workflows with Health IT

Learning objectives

- Explain workflows in the context of Health Informatics & Digital Health.
- Apply Health IT to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

workflows, Health IT, healthcare

Retrieval prompt

In Health Informatics & Digital Health, which evidence best supports a workflows result produced with Health IT?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

FHIR: Quality and safety

Apply FHIR through validation, privacy, human oversight.

LESSON 07 / 26 MIN

Validation with EHR

Learning objectives

- Explain validation in the context of Health Informatics & Digital Health.
- Apply EHR to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

validation, EHR, healthcare

Retrieval prompt

In Health Informatics & Digital Health, which evidence best supports a validation result produced with EHR?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 08 / 30 MIN

Privacy with FHIR

Learning objectives

- Explain privacy in the context of Health Informatics & Digital Health.
- Apply FHIR to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

privacy, FHIR, healthcare

Retrieval prompt

In Health Informatics & Digital Health, which evidence best supports a privacy result produced with FHIR?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 09 / 18 MIN

Human oversight with HL7

Learning objectives

- Explain human oversight in the context of Health Informatics & Digital Health.
- Apply HL7 to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

human oversight, HL7, healthcare

Retrieval prompt

In Health Informatics & Digital Health, which evidence best supports a human oversight result produced with HL7?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

HL7: Applied evaluation

Apply HL7 through implementation, monitoring, equity.

LESSON 10 / 22 MIN**Implementation with Telemedicine**

Learning objectives	• Explain implementation in the context of Health Informatics & Digital Health. • Apply Telemedicine to a bounded practical task. • Evaluate the result using explicit quality criteria.
Key terms	implementation, Telemedicine, healthcare
Retrieval prompt	In Health Informatics & Digital Health, which evidence best supports a implementation result produced with Telemedicine?
Evidence to keep	☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 11 / 26 MIN**Monitoring with Health IT**

Learning objectives	• Explain monitoring in the context of Health Informatics & Digital Health. • Apply Health IT to a bounded practical task. • Evaluate the result using explicit quality criteria.
Key terms	monitoring, Health IT, healthcare
Retrieval prompt	In Health Informatics & Digital Health, which evidence best supports a monitoring result produced with Health IT?
Evidence to keep	☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 12 / 30 MIN**Equity with EHR**

Learning objectives	• Explain equity in the context of Health Informatics & Digital Health. • Apply EHR to a bounded practical task. • Evaluate the result using explicit quality criteria.
Key terms	equity, EHR, healthcare
Retrieval prompt	In Health Informatics & Digital Health, which evidence best supports a equity result produced with EHR?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable Health Informatics & Digital Health project using Health IT, EHR, FHIR.

DELIVERABLE	A working Health Informatics & Digital Health artefact plus an evidence-based self-review.
TOOLS	A suitable Health IT environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using Health IT.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

RESPONSIBLE PRACTICE

This material is educational and does not provide medical advice or replace qualified clinical judgement.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER

State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS

List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE

Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION

Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

Global strategy on digital health 2020-2027

World Health Organization - reviewed 2026-08-21

<https://www.who.int/publications/i/item/9789240116870>

FHIR specification

HL7 International - reviewed 2026-08-21

<https://hl7.org/fhir/>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Health IT scope.